HKN ECE 310 Review Session

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Logistics

Slides and recording can be found on the HKN website (hkn.illinois.edu)

This session is recorded.

Exam reminders:

- 20% of Final Grade
- Wednesday, 9/30, 7:00-9:00pm
- One 8.5 x 11” sheet of handwritten notes (double-sided) allowed

Overview

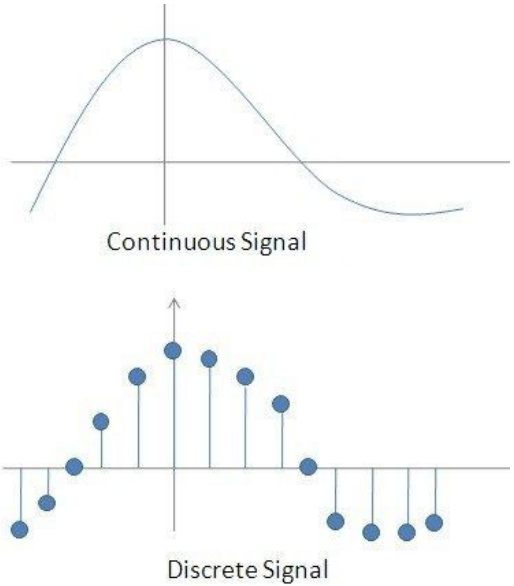
Topics

- CT and DT Signals
- LTI and LTIC Systems
- Impulse Response
- Convolution
- Z-transforms
- Stability, Causality, etc
- Multiple Systems

CT and DT Signals

- Discrete-Time (DT) signals are really just Continuous-Time (CT) signals, but sampled at some regular interval

- $x(t)$ → Continuous-Time (t real-valued)
- $x[n]$ → Discrete-Time (n → integer)



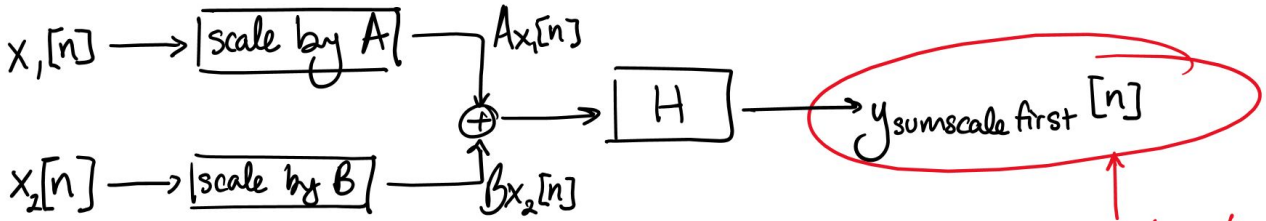
LTI and LTIC Systems

Given a system H which receives an input x[n] and produces an output y[n], we can check for three properties:

- **Linearity** — H is *linear* if:
  - Given $x_1[n] \rightarrow y_1[n]$, and given $x_2[n] \rightarrow y_2[n]$, then $ax_1[n] + bx_2[n] \rightarrow ay_1[n] + by_2[n]$
  - Any scaling and summing of original inputs is equivalently reflected in the outputs
- **Time Invariance** — H is *time-invariant* (AKA shift-invariant) if:
  - Given $x[n] \rightarrow y[n]$, then $x[n-k] \rightarrow y[n-k]$
  - Any delay in the original input is equivalently reflected in the output
- **Causality** — H is *causal* if:
  - The current output $y[n]$ does not depend on future input/output samples *for all values of n*

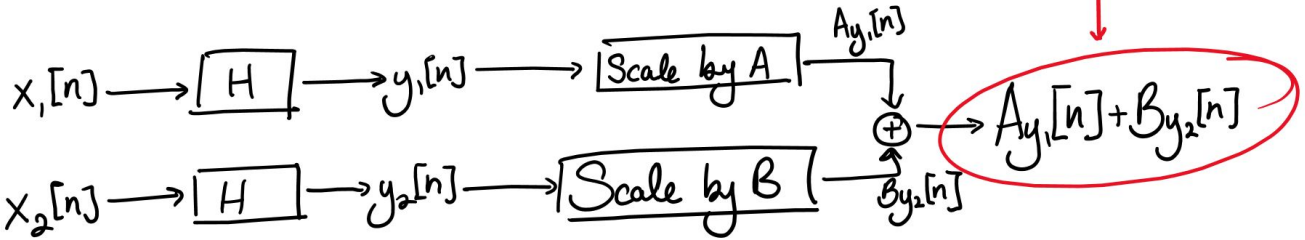
Linearity Visualized

Apply and Feed



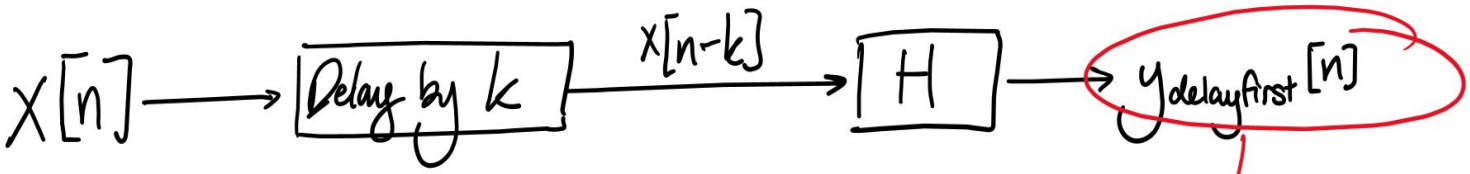
Same iff H is Linear

Feed and Apply



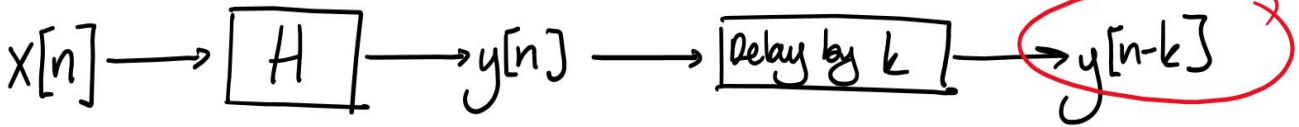
Time Invariance Visualized

Apply and Feed



Same iff H is TI

Feed and Apply



General Approach on Proving L/TI

1. Feed and apply
   a. Define input/output pairs to the system
      i. $x[n] \rightarrow y[n]$
   b. Apply transformation to the outputs
2. Apply and feed
   a. Define a new function applying whatever variant
      i. Testing time invariance: $g[n] = x[n-k]$
      ii. Testing linearity: $g[n] = ax_1[n] + bx_2[n]$
      iii. (Not strictly necessary, but it's easy to get lost…)
   b. Feed this into the system determine output $y_g[n]$
3. Compare the results from steps 1 & 2

Examples: Proving LTI via a system's equations

- Prove, or disprove, that $y[n] = \frac{x[2n]}{n^2}$ is linear.
- Prove, or disprove, that $y[n] = \frac{x[n-\alpha]}{x[n-\beta]}$ is time invariant.
- Prove, or disprove, that $y[n] = x[n] + x[2024]$ is time invariant.

Causality of Systems via their Equations

- Suppose we have some equation describing some system
  - $y[n] = x[n-4] + y[n-1] + n^2 x[n]$
- “Current outputs cannot depend on future inputs or outputs”
- A more formal definition:
  - Given an equation for $y[n]$, the time index of every input/output term on the right side of the equation must be less than or equal to n, for all values of n.

Justifying Causality of Systems via their Equations

For each system, determine whether it is causal or not and justify:

1. $y[n] = x[n] + x[n-1] + x[n-2] + \ldots$
2. $y[n] = x[n] + y[n-1] + x[n-200] + x\left[\left\lfloor \frac{n}{2} \right\rfloor\right]$
3. $y[n] = \cos(2024n)$
4. $y[n] = |x[n]|$
5. $y[n] = x[|n|]$
6. $y[n] - y[n-3] = x[n+2]$

BIBO Stability of Systems via their Equations

- A system is **BIBO stable** if and only if every bounded input $x[n]$ produces a bounded output $y[n]$.
  - That is, if $|x[n]| < \boldsymbol{\beta}$ for all n, then $|y[n]| < \boldsymbol{\alpha}$ for all n, where $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$ are finite constants.
  - “Bounded output” must be well-defined output

Justifying BIBO Stability of Systems via their Equations

Determine whether the following sequences are BIBO stable or not. If not, provide a bounded input that causes the input to be unbounded.

1. $y[n] = 20x[n]$
2. $y[n] = \dfrac{(½)^n x[n]}{x[310]}$
3. $y[n] = \dfrac{1}{n^2} + \dfrac{x[n]}{1-(x[n]-5)^2}$
4. $y[n] = x[n] + x[n-1] + y[n-1]$
5. $y[n] = x[n](-1)^n - 2y[n-1]$

Impulse Response

A system's **impulse response** is the output you get when you pass in the unit impulse $\delta[n]$ as an input.

Let $x[n]$ be the input to an **LTI** system with impulse response $h[n]$. Then, the system output $y[n]$ is given by

$$y[n] = \sum_{m=-\infty}^{+\infty} x[m]h[n-m]$$

which can be written as $y[n] = x[n] * h[n]$.

Properties of the Impulse Response

For an LTI system with impulse response $h[n]$:

- The system is **causal** if $h[n] = 0,\ n < 0$
  - i.e. $h[n]$ is right-sided *and* begins at or after n=0
  - If there is some n < 0 for which $h[n] \neq 0$, the system is **non-causal**
- The system is **BIBO stable** if $h[n]$ is absolutely summable, or

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Convolution

Convolving by hand can be done in two main ways:

1. Convolution sum
2. Shift-and-overlap method / Matrix method

$$y[n] = \sum_{m=-\infty}^{+\infty} x[m]h[n-m]$$

Convolution sum

When should I use each one?

- Both signals finite-length → **Shift-and-overlap**
- Both signals infinite-length → **Convolution sum**
- One finite-length, one infinite-length → **Use convolution properties** (more on those later)

Which of the following make sense?

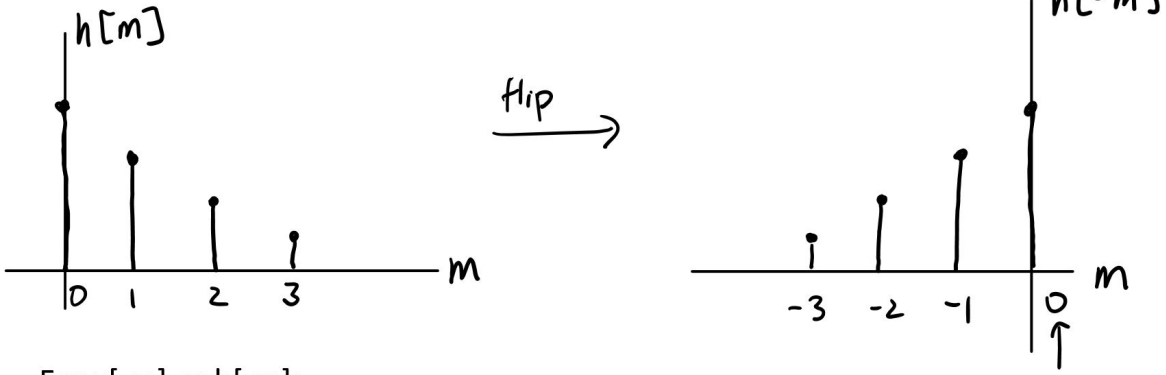
Let x[n] be some arbitrary signal. Let h[n] be the impulse response of some linear system.

1. x[n] is linear.
2. x[n] is shift-invariant.
3. x[n] is causal.
4. x[n] is BIBO-stable.
5. x[n] is unbounded.

1. h[n] is linear.
2. h[n] is shift-invariant.
3. h[n] is causal.
4. h[n] is BIBO-stable.
5. h[n] is unbounded.

Graphical Method

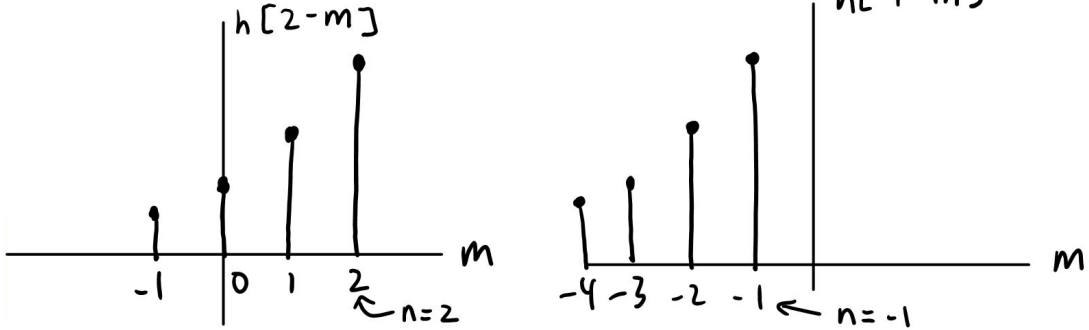
1. Flip either x[n] or h[n] and find as a function of m rather than n



For x[-m] or h[-m]:

- n > 0 → shift right
- n < 0 → shift left

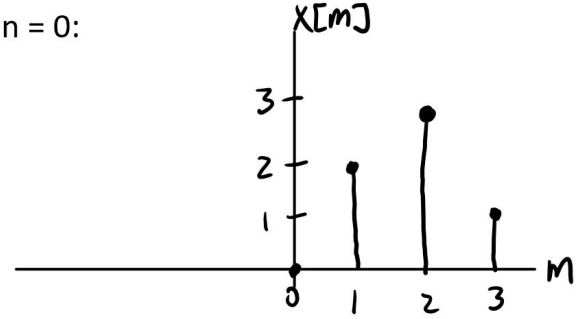
This m = 0 point is equivalent to the n value during convolution

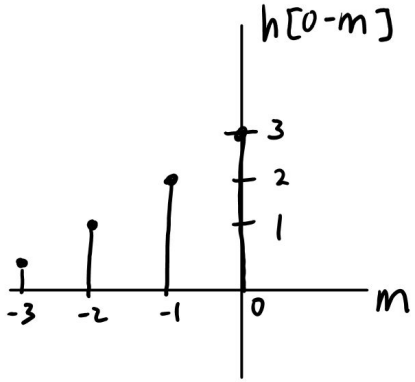


Graphical Method (cont)

2. Pass the flipped signal, x[n-m] or h[n-m] from left to right and find the convolution result $y[n] = \sum_{m=-\infty}^{+\infty} x[m]h[n-m]$ or $y[n] = \sum_{m=-\infty}^{+\infty} h[m]x[n-m]$ as follows:

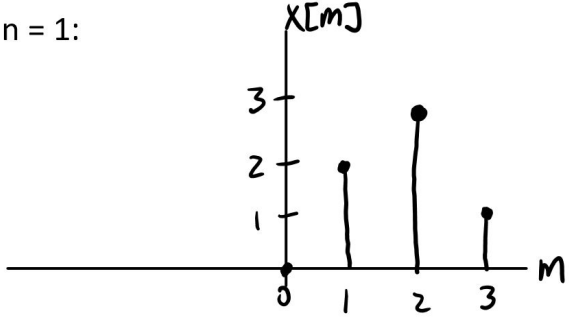
n = 0:

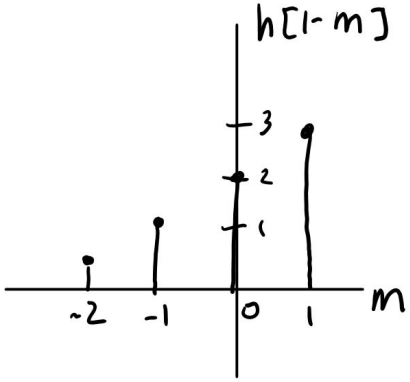




y[0] = 0 * 3 = 0

n = 1:

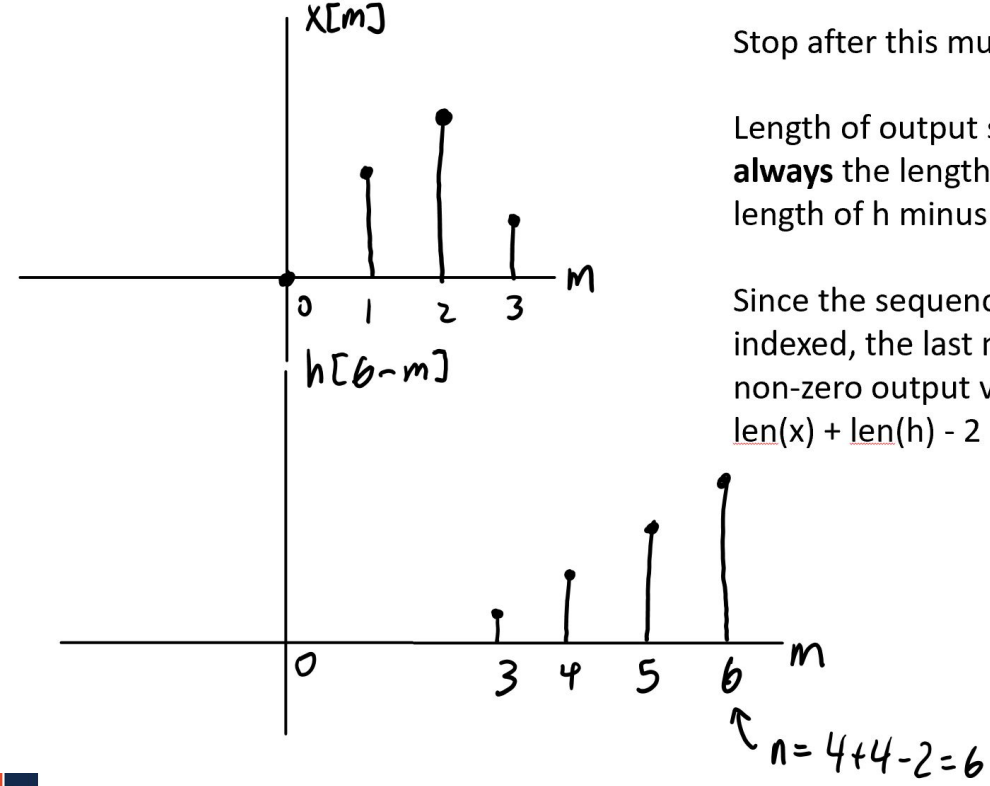




y[1] = 0 * 2 + 2 * 3 = 6

Graphical Method (cont)

3. Continue until you have found a number of points equal to the length of x + the length of h minus 1



Stop after this multiplication!

Length of output sequence is **always** the length of x + the length of h minus 1!

Since the sequences are zero-indexed, the last n that gives a non-zero output value should be len(x) + len(h) - 2

Sequence Method (cont)

3. Continue until you have found a number of points equal to the length of x + the length of h minus 1

n = 6:

x[m] = [-1, 0, 1, 2]

h[6-m] = [0, 0, 0, 0, 1, 3, 2]

can be omitted
(for a metaphor to
the graphical method)

y[6] = 2 * 0 = 0

Stop after this multiplication!

Length of output sequence is **always** the length of x + the length of h minus 1!

Since the sequences are zero-indexed, the last n that gives a non-zero output value should be len(x) + len(h) - 2

Shift-and-Overlap Method

Given two sequences x[n] and h[n]:

- x[n] = [-1, 0, 1, 2]
  ↑
- h[n] = [2, 3, 1, 0]
  ↑

1. First flip one of the two sequences, and obtain both as a function of m

- x[m] = [-1, 0, 1, 2]
  ↑
- h[-m] = [0, 1, 3, 2]
  ↑

Shift-and-Overlap Method (cont.)

2. Shift the flipped sequence from left to right, multiplying overlapping elements and adding the products together for each n

The starting index of your output $y[n]$ will simply be the starting index of $x[n]$ + the starting index of $h[n]$

m = 0

x[m] = [-1, 0, 1, 2]

n = 0: h[0-m] = [0, 1, 3, 2] y[0] = -1 * 2 = -2

n = 1: h[1-m] = [0, 1, 3, 2] y[1] = -1 * 3 + 0 * 2 = -3

And so on…

Shift-and-Overlap Method (cont.)

3. Continue until you've found a number of points equal to the length of $x[n]$ + the length of $h[n]$ - 1

		m = 0	
	x[m] =	[-1, 0, 1, 2]	
n = 6:	h[6-m] =	[0, 1, 3, 2]	y[6] = 0 * 2 = 0

Length of $y[n]$: 4 + 4 - 1 = **7**

Matrix Method

Turn one of your signals to a column vector, the other into a matrix. Multiply -> Profit

x[n] = [-1, 0, 1, 2]
h[n] = [2, 3, 1, 0]

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & 2 & 0 & 0 \\ 1 & 3 & 2 & 0 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ 1 \\ 7 \\ 7 \\ 2 \end{bmatrix}$$

^ we truncated h[n] to be [2, 3, 1] above

Convolution Properties

Just like multiplication, convolution is:

- Commutative
- Associative
- Distributive

$$x[n] * h_1[n] * h_2[n] = x[n] * h_2[n] * h_1[n]$$

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n]).$$

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n].$$

Other properties:

- Any signal convolved with $\delta[n]$ is the exact same signal $x[n] * \delta[n] = x[n]$
- Convolution is linear and time-invariant

$$h[n] * a\delta[n-k] = ah[n-k]$$

Impulse response and causality

Suppose we have an LTI system with impulse response $h[n]$.

- The system is causal (the system is LTIC) implies: $h[n] = 0$ for $n < 0$.
  - Signals for which $x[n] = 0$ for $n < 0$ are called “causal signals” for this reason.
- The system is non-causal implies $h[n] \neq 0$ for $n < 0$

Z-transforms

- The z-transform of a signal y[n] is defined as:

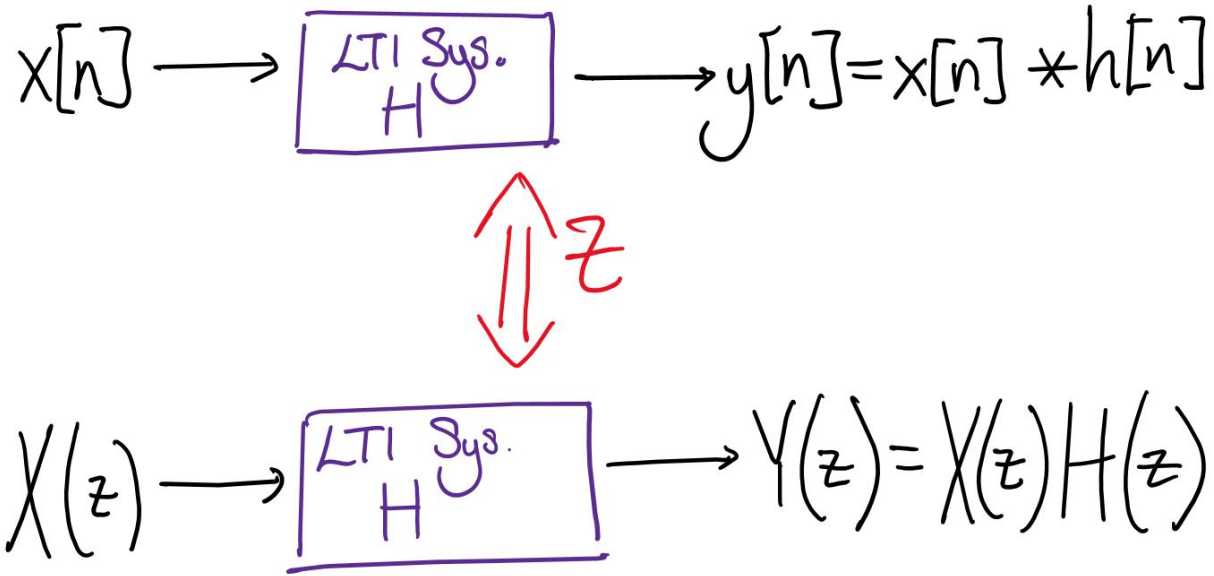
$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]z^{-n}$$

- A Z-transform is uniquely determined by TWO things:
  - Expression for transform, F(z)
  - Region of Convergence (ROC), range of |z|
    - Continuous region
    - Does not include poles
- Inverse z-transform calculated via inspection and/or partial fraction decomposition

Z-transform Properties

- Important properties:
  - Linearity
  - Time-shifting (AKA delay)
  - Convolution
    - $x[n] * h[n] \leftrightarrow X(z)H(z)$
- Each property may affect the ROC when applied
  - ex: Time reversal causes |z| > a to become |1 / z| > a
- Tips:
  - Look out for opportunities to use properties to go fast on the exam!
  - When the signal doesn't neatly fit a transform pair, try using the original summation definition of the z-transform
  - Complete list of properties and transform pairs can be found on the website under "Resources"

Convolution Theorem Visualized



Z-transforms: Example

$$x[n] = 5^n u[n-2] + (0.2)^n \delta[n+1]$$

$$x[n] = 5^2(5^{n-2})u[n-2] + (0.2)^n \delta[n+1]$$

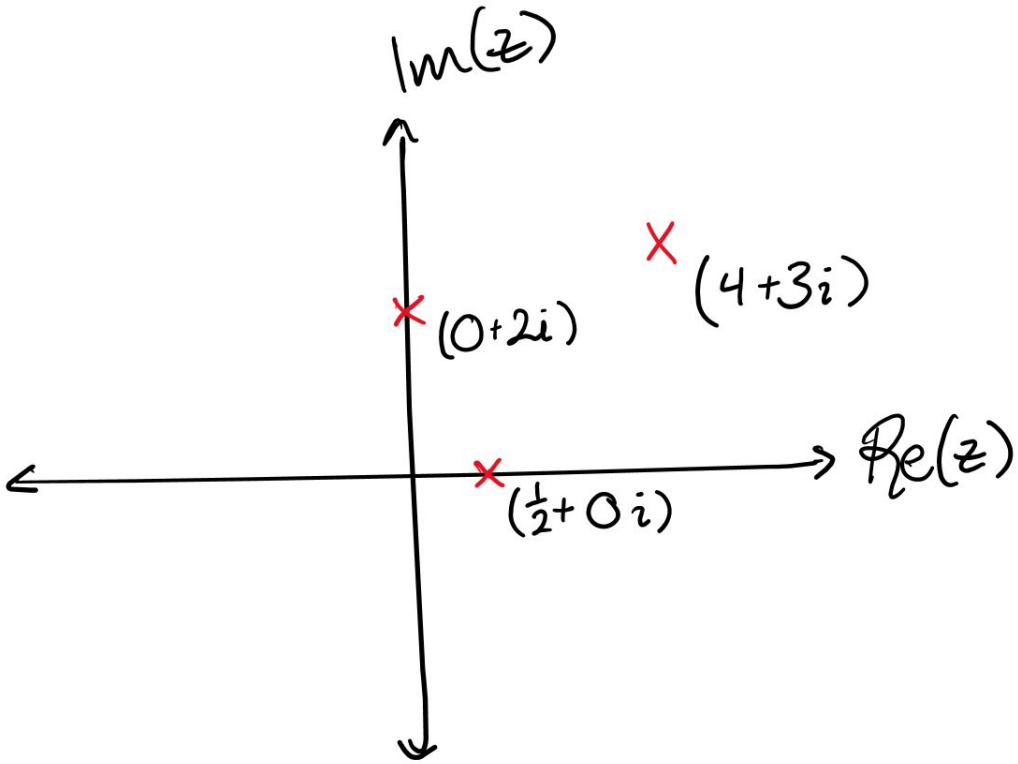
$$Z \downarrow$$

$$X(z) = \frac{5^2 z^-2}{1-5z^-1} + 5z$$

ROC: 5 < |z| < +∞

Concept check: Possible ROCs

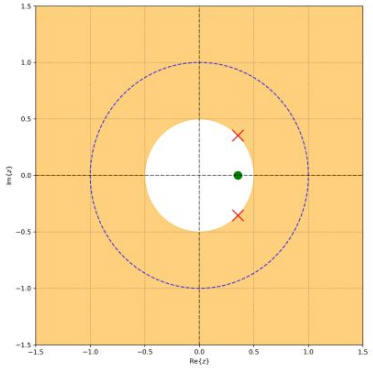
Here is a plot of the poles of a z-transform. What are all possible ROCs?

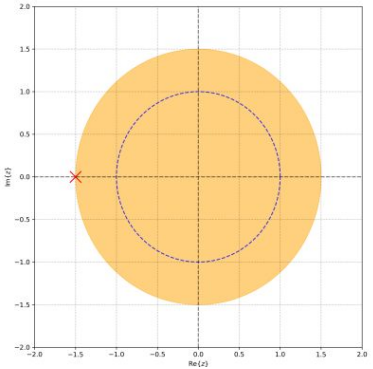


Signals and their ROCs – General Patterns

Suppose we have $x[n] \leftrightarrow X(z)$.

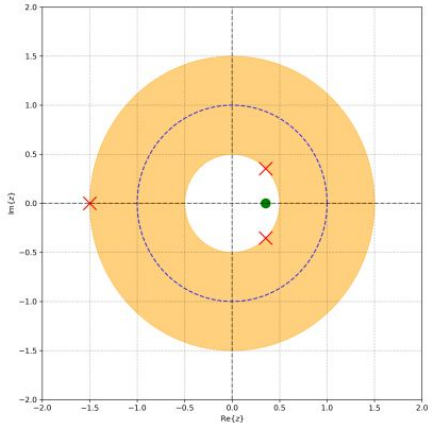
- Right-sided signal ($x[n] = 0$ for $n < k$ for some k):
  - Example: $x[n] = u[n-4]3^{n-4}$
  - ROC: $|z| > r$ for some r
  - Hint: Causal impulse responses fall under this category.
- Left-sided signal ($x[n] = 0$ for $n > k$ for some k):
  - Example: $x[n] = u[-(n+1)]4^{n+1}$
  - ROC: $|z| < r$
  - Hint: Anticausal impulse responses fall under this category (not to be confused with *non-causal*)





Signals and their ROCs – General Patterns

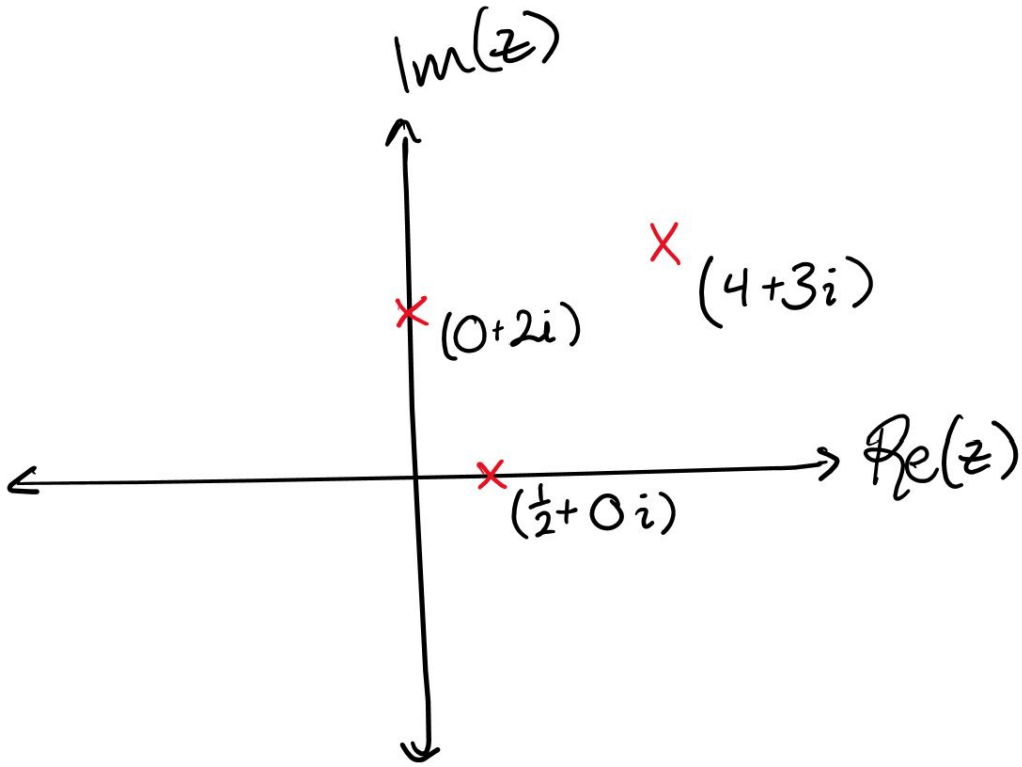
- Finite signal (both right-sided and left-sided)
  - ROC: all |z| except potentially |z| = 0 and/or |z| = infinity
- Infinite signal in both directions (neither left-sided nor right-sided)
  - ROC: a < |z| < b for some finite constants a, b



Concept check: Signals and their ROCs

Here is the same plot from before.

Which ROCs result in a double-sided signal? Left-sided? Right-sided?



Example: Inverse Z-transform

Given

$$H(z) = \frac{z^{-1}}{(1-\frac{1}{4}z^{-1})(1-\frac{1}{2}z^{-1})}$$

and that $H(z)$ represents the transfer function for a causal system, find the impulse response.

Relationship between LTI systems and Transfer Function

A system is LTI if and only if...

1. Its transfer function H(z), the z-transform of the impulse response, exists
2. H(z) is equal to the ratio of the z-transform of the output to the z-transform of the input.

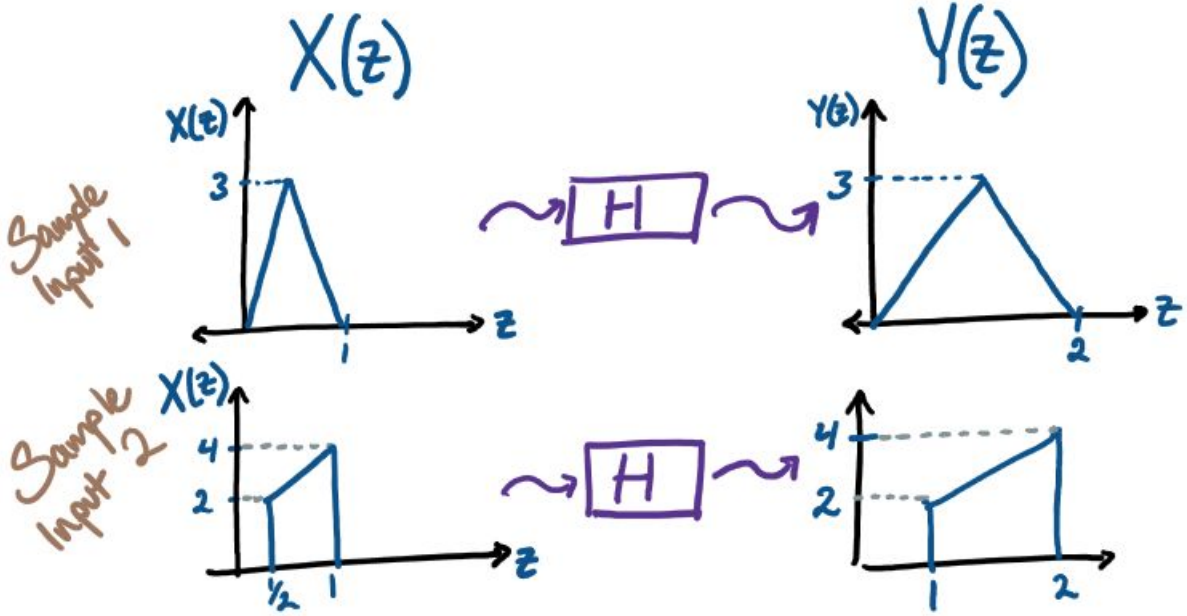
$$H(z) = \frac{Y_a(z)}{X_a(z)} = \frac{Y_b(z)}{X_b(z)}$$

- This ratio is the same for every input/output pair.

Checking for LTI in the Z-domain

We are given a system H. We send two different inputs, $x_1[n] \leftrightarrow X_1(z)$and $x_2[n] \leftrightarrow X_2(z)$ and obtain outputs $y_1[n]$and $y_2[n]$. Is it possible that this system is LTI, or is it impossible?

$Y(z) = X\left(\frac{z}{2}\right)$



[a] Possibly LTI

[b] Absolutely not LTI

Conceptual Check

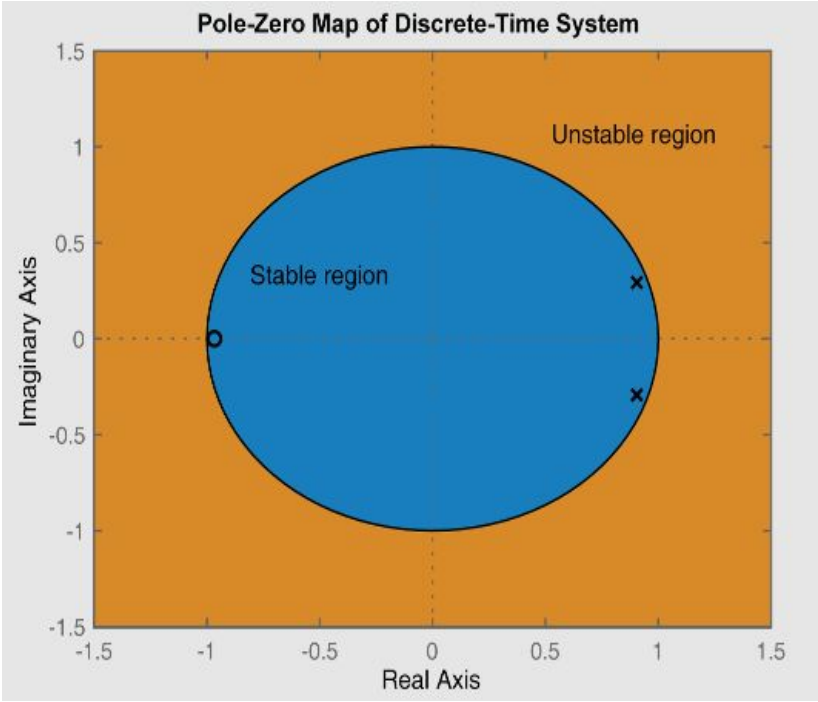
Suppose we know a system is LTI. Suppose that some finite-length input $x_a[n] \leftrightarrow X_a(z)$ has output $y_a[n] \leftrightarrow Y_a(z)$.

- True/False: If we know that for some z_0, $Y_a(z_0) = 0$, but $X_a(z_0) \neq 0$, then we know that $Y_{\text{any input}}(z_0) = 0$.

- True/False: Consider some value z_0. Knowing $X_a(z_0) = 0$ does not give us information about the value of $Y_a(z_0)$.

Marginal Stability

- An LTI system is **marginally stable** if none of its poles have a radius greater than 1, and if it has one or more distinct poles with radius 1.
- For a system in general, it is marginally stable if the ROC borders |z| = 1 but doesn't contain it.
- In this class, marginally stable = **unstable**.



Source: https://in.mathworks.com/help/ident/ref/lti.pzmap.html

Stability and Causality of LTI systems

Given an LTI system with impulse response $h[n] \leftrightarrow H(z)$:

- H is BIBO stable if and only if the ROC of the z-transform $H(z)$ includes the unit circle.
- H is causal implies that the ROC of $H(z)$ has the form |z| > r for some r.

BIBO Stability of LTI systems

- Checking for BIBO stability of LTI systems

1. Absolute summability of impulse response
   a. i.e. $\sum_{n=-\infty}^{+\infty}|h[n]| < \infty$
2. Location of poles and ROC of the transfer function
   a. ROC must include unit circle (poles + causality)
3. System definition (proof by counterexample)
   a. Find a bounded input that produces an unbounded output
   b. Only method that's also applicable to non-LTI systems

Example

$$H(z) = \frac{1 - 2z^{-2}}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$$

- Find the LCCDE for the system.
- Is the system BIBO stable?

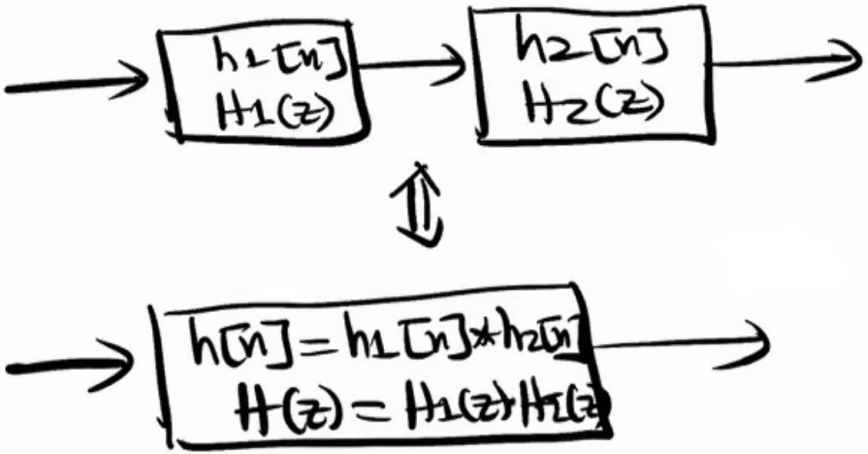
A semi-mechanical way to find BIBO unstable pairs

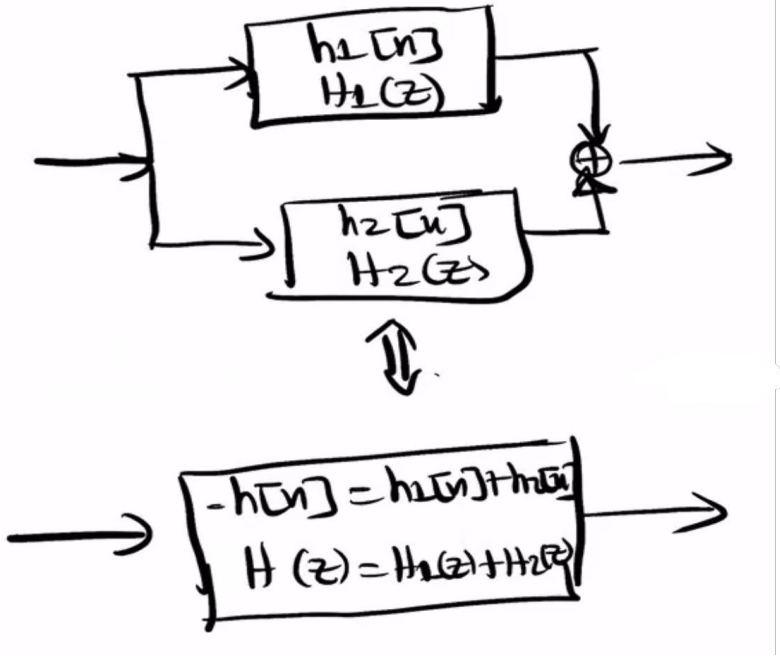
1 Finding Bounded output and unbounded output pairs

1. Rewrite the denominator in terms of $(1 - kz^{-1})$ factors exclusively
2. If $\exists k$ where $|k| > 1$, then $\delta[n]$ makes it unbounded
3. If $\exists k$ where $k = 1$, then $u[n]$ makes it unbounded
4. If $\exists k$ where $|k| = 1$ but $k \neq 1$, then $k = e^{j\omega n}$. Then $\cos(\omega n)u[n]$ makes it unbounded

Multiple Systems

- We can combine systems in **serial** or **parallel** connections.





Bonus: ECE 310 MT1 any% speedrun tactics

- Time is a concern
- PFE: change of variable / direct PFE
- Polynomial division: Google “synthetic division”
  - Useful if the denominator is factorable into single-root terms
- Tables: Use the properties and table pairs whenever you can!
  - Z-transform pairs/properties, DTFT pairs/properties: learn them by heart, or find a place on your reference sheet to put them.

Exam Structure

~8-9 Questions

- Conceptual T/F
- LTIC T/F
- Convolution (given input/outputs)
- Calculating Z-transform
- Calculating System Response (given transfer function and signal input)
- Finding BIBO Stability

Conceptual Question

- If a system is BIBO stable then it must be causal. T/F
- A BIBO stable system with a transfer function $H(z) = \frac{1}{1 - 2z^{-1}}$ must be non-causal T/F
- For a given system with x[n] as the signal input, y[n] as the signal output, and h[n] as the impulse response, $y[n] = x[n] * h[n]$ ALWAYS describes their relationship T/F
- If a system has a right-sided unit pulse response, then it must be causal. T/F

LTIC and BIBO Stability

	Linear	Time-invariant	Causal	Stable
$y[n] = (x * u)[n]$				
$y[n] = \frac{x[n]}{x[2]}$				

Convolution

1) x[n] = {1, 2, -3, -4, 5}; h[n] = {-3, 1, 2} (x[0] = -3 and h[0] = 1)

1) x[n] = 0.5 ^ n * u[n]; h[n] = (n - 1) * (u[n+1] - u[n-2])

Convolution given output

Suppose y[n] = x[n] * h[n]. Given y[n] = 27*u[n-1] - 3*u[n+1] and x[n] = {-1, 0, 9} where x[0] = -1, what is h[n]?

LCCDE

Consider the following LCCDE: $y[n] = \frac{8}{15}y[n-1] - \frac{1}{15}y[n-2] + x[n]$

Determine the following

1) Transfer Function
2) ROC
3) Zeros and Poles
4) Impulse Response
5) BIBO Stability
6) If x[n] = {-5, 1} where x[0] = -5, find y[n]

Thank you!

Good luck on Midterm 1!

Let us know if we were helpful:

